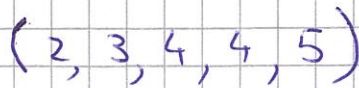


4/2/25

Q1


$$(1, 2, 3, 5)$$

Q 2

AA DD

BB

AA D F A

$AA \triangle B$

AAA

AAA B

AA Δ A

AASD

A A D F A

 $AA \supset B$

AAA A

AAA B

A A D A

$$S_2, 2$$
 $S \geq 1$

BB

STOP

$\Rightarrow$ same set point
To $S = 3$

AAⓅB ← pivot

AADFA

AADB

AAAA

AAA B

AADA

BB

$S_{<,3}$ { AAAA
AAA B
AADD
AADB
AADA
BB

AADFA (pick this as pivot)

$S_{=,4}$ { AAAⓅ ← pivot
AAA B

$S_{=,5}$ AAAA

AAAA

$S_{>,4}$ AAA B

AAA B

AADⓅA ← pivot

AADD

$S_{=,4}$ AADB

AADA

BB

$S_{<,4}$ { AADⓅ ← pivot
AADB
AADA

$S_{=,5}$ { AADFA

BB

$S_{<,4}$ { AADⓅ ← pivot
AADA

$S_{=,5}$ { AADD

AADFA

BB

AAAA

AAA B

$S_{<,4}$ { AADA

$S_{=,5}$ { AADB

AADD

AADFA

BB

sorted!

Q3 Apply Elias-Fano on the following sequence

$$S = (1, 2, 3, 6, 8, 10, 11, 15, 19, 23, 31)$$

$$u = \max S + 1 = 32$$

$$b = \log_2 32 = 5$$

$$l = \left\lceil \log_2 \frac{u}{n} \right\rceil = \left\lceil \log_2 \frac{32}{11} \right\rceil = 2 \text{ bits} \Rightarrow u = 3 \text{ bits}$$

1	000	01
2	000	10
3	000	11
6	001	10
8	010	00
10	010	10
11	010	11
15	011	11
19	100	11
23	101	11
31	111	11

↑ array L

$$H = \frac{1110}{0} \frac{10110}{1} \frac{10}{2} \frac{10}{3} \frac{10}{4} \frac{10}{5} \frac{0}{6} \frac{10}{7}$$

Access (5) → pick 5th pair of bits in L → 00

$$y = \text{select}_1(5, H) - 5 = 7 - 5 = 2$$

represented in 3 bits → 010

Hence, result is 01000 = 8

Next $q \in Q(9) \Rightarrow$ we represent 9 in 5 bits → $\frac{01001}{\text{high low}}$

$$y = \text{select}_0(010) + 1 \rightarrow \text{select}_0(2) + 1 = 7$$

$$\text{rank}_1(7) = 5 \quad (\text{oppose } [\text{select}_0(7)] \rightarrow (y - 2) = 5)$$

010

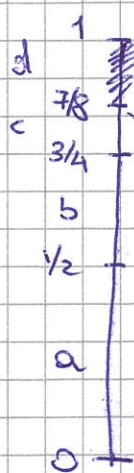
Access (5) → 8 < 9

Access (6) → 10 > 9 stop & return 10

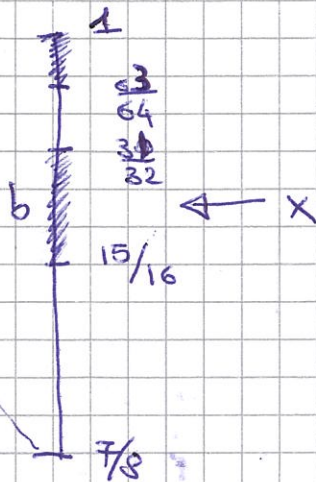
SCAN

Q4

$$x = \underset{1/2}{111101} \rightarrow \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} = \frac{32+16+8+4+1}{64} = \frac{61}{64}$$



$S = 1$



$S = 1/8$

$T = db$

Q5

$B[1, 2n]$ binary encoding of a binary Tree
 $L[1, n]$ node labels in $\{A, B, C, D\}$

Design an algorithm that checks the existence of an edge (A, B)

① We scan for B, and check whether the parent is A

for $(i = 2; i < 2n + 1; i++)$

if $(B[i] == 1) \text{ \&\& } (L[\text{rank}_1(i)] == 'B')$ {

$J = \left\lfloor \frac{i}{2} \right\rfloor$ $\rightarrow$ select, $(\text{rank}_1(i))$ is the node 'i' does exist

if $L[J] = 'A'$ return TRUE

return FALSE

② Another solution consists of finding all positions in L where the labels A and B occur, and then check that for them

holds : $\exists p_A, \exists p_B$ s.t. $B[2p_A] == 1 \text{ \&\& } L[\text{rank}_1(2p_A)] == 'B'$
 or $B[2p_A + 1] == 1 \text{ \&\& } L[\text{rank}_1(2p_A + 1)] == 'B'$